

AI-BASED DYNAMIC SKINCARE RECOMMENDER FOR WOMEN 40+

A Minor Project Report

Submitted in partial fulfillment of requirement of the

Degree of

**BACHELOR OF TECHNOLOGY in COMPUTER
SCIENCE & ENGINEERING**

BY

Chhavi Raghuvanshi

Under the Guidance of

Prof. Amrata Gupta | Prof. Laxmi Kag



**Department of Computer Science & Engineering
Faculty of Engineering
MEDICAPS UNIVERSITY, INDORE- 453331**

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Report Approval

The project work “AI-BASED DYNAMIC SKINCARE RECOMMENDER FOR WOMEN 40+” is hereby approved as a creditable study of an engineering/computer application subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.

It is to be understood that by this approval the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion drawn therein; but approve the “Project Report” only for the purpose for which it has been submitted.

Internal Examiner

Name:

Designation

Affiliation

External Examiner

Name:

Designation

Affiliation

Declaration

We hereby declare that the project entitled “AI-BASED DYNAMIC SKINCARE RECOMMENDER FOR WOMEN 40+” submitted in partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science & Engineering completed under the supervision of **Prof. Amrata Gupta Prof. Laxmi Kag**, Faculty of Engineering, Mediacaps University Indore is an authentic work.

Further, we declare that the content of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for the award of any degree or diploma.

Chhavi Raghuvanshi

Certificate

We, **Prof. Amrata Gupta and Prof. Laxmi Kag** certify that the project entitled AI-BASED DYNAMIC SKINCARE RECOMMENDER FOR WOMEN 40+ submitted in partial fulfillment for the award of the degree of Bachelor of Technology by **Chhavi Raghuvanshi** is the record carried out by them under our guidance and that the work has not formed the basis of award of any other degree elsewhere.

Prof. Amrata Gupta ,

Department of Computer Science &
Engineering

Medicaps University, Indore

Prof. Laxmi Kag

Department of Computer Science &
Engineering

Medicaps University, Indore

Prof.(Dr) Kailash Chandra Bandhu

Head of the Department

Computer Science & Engineering

Medicaps University, Indore

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Chhavi Raghuvanshi

B.Tech. III Year (T)
Department of Computer Science & Engineering
Faculty of Engineering
Medicaps University, Indore

Abstract

The **AI-Based Dynamic Skincare Recommender for Women 40+** is an intelligent system designed to provide personalized skincare recommendations based on individual skin concerns. The project focuses specifically on women above the age of 40, who often experience skin-related issues due to hormonal changes, aging, and environmental factors.

The system utilizes a dynamic search-based approach to analyze user inputs such as dryness, pigmentation, wrinkles, and sensitivity, and maps these concerns to suitable skincare ingredients and products. By leveraging structured data and rule-based recommendation logic, the system ensures accurate and relevant suggestions tailored to each user's needs.

The primary objective of this system is to assist users in making informed skincare decisions without requiring expert knowledge. It reduces the time and effort involved in selecting appropriate products and provides a more targeted and efficient solution compared to generalized skincare recommendations available in the market.

The platform is designed with a user-friendly interface and efficient backend processing, ensuring quick response time and ease of use. It also supports scalability for future enhancements such as AI-driven skin analysis and integration with real-time data sources.

Overall, this project demonstrates how intelligent systems can be applied in the skincare domain to improve personalization, accessibility, and decision-making for a specific and often underserved demographic group.

Keywords

Artificial Intelligence, Skincare Recommendation System, Dynamic Search, Personalized Recommendations, Skin Concerns, Ingredient Mapping, Decision, Support System, Women 40+ Skincare

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Abbreviations

Abbreviation	Full Form
API	Application Programming Interface
DFD	Data Flow Diagram
ER	Entity-Relationship
GUI	Graphical User Interface
HTTP/HTTPS	Hypertext Transfer Protocol / Secure
IDE	Integrated Development Environment
JSON	JavaScript Object Notation
JWT	JSON Web Token
NLP	Natural Language Processing
NMT	Neural Machine Translation
OOP	Object-Oriented Programming
ORM	Object-Relational Mapping
SRS	Software Requirements Specification
SDS	Software Design Specification
UI	User Interface
UML	Unified Modeling Language
URL	Uniform Resource Locator
WCAG	Web Content Accessibility Guidelines

Notations & Symbols

Symbol / Notation	Meaning / Usage
→	Data flow direction in DFDs and sequence diagrams
[]	Optional element or array notation in data structures
{ }	Set or group of related elements
FR-n	Functional Requirement identifier (e.g., FR-1, FR-2)
NFR-n	Non-Functional Requirement identifier
PK	Primary Key in database table design
FK	Foreign Key in database table design
1NF / 2NF / 3NF	First / Second / Third Normal Form (database normalization)
VARCHAR(n)	Variable-length character string of max n characters
INT	Integer data type in database
TEXT	Long-form text data type in database
DATETIME	Date and time combined data type
BOOLEAN	True/False binary data type
≤	Less than or equal to (used in performance thresholds)

CHAPTER 1: INTRODUCTION

1.1 Introduction

In recent years, the global skincare industry has witnessed rapid growth, driven by increased awareness of personal health, aging, and lifestyle-related skin conditions. Among various demographic groups, women aged 40 and above represent a particularly important yet underserved segment. This group experiences significant physiological and hormonal changes, especially during and after menopause, which directly impact skin health.

These biological changes lead to a variety of skin concerns such as dryness, reduced elasticity, wrinkles, pigmentation, and increased sensitivity. The decline in collagen production, reduced sebum secretion, and slower skin cell regeneration contribute to these issues. Despite the growing demand for targeted solutions, most skincare products and recommendations available in the market remain generalized and are not tailored to the specific needs of this age group.

Traditional skincare selection methods rely heavily on trial-and-error approaches, influencer trends, or generalized advice, which often results in ineffective or unsuitable product choices. Furthermore, access to dermatological consultation is limited for many users due to cost, time constraints, or geographical barriers.

The emergence of Artificial Intelligence (AI) and data-driven systems has created new opportunities to address these challenges. AI-based recommendation systems can analyze user-specific inputs and provide personalized suggestions with higher accuracy and efficiency. Such systems have already demonstrated success in domains like e-commerce, healthcare, and content streaming.

The **AI-Based Dynamic Skincare Recommender for Women 40+** is developed in this context as a practical, user-centric solution. The system utilizes a dynamic search mechanism to map user-selected skin concerns to relevant skincare ingredients and recommendations. Instead of relying on static suggestions, the system dynamically processes user input and retrieves personalized results from a structured database.

By focusing specifically on women aged 40+, the system addresses a critical gap in the skincare domain. It provides a simple, accessible, and efficient platform that enables users to make informed skincare decisions without requiring expert-level knowledge. The project demonstrates how intelligent systems can be effectively applied to improve personalization and accessibility in everyday health and wellness applications.

1.2 Literature Review

The development of recommendation systems has evolved significantly over the past two decades, transitioning from simple rule-based systems to advanced AI-driven models. This section reviews existing approaches in skincare recommendation and identifies the gaps addressed by this project.

Early recommendation systems were primarily rule-based, where predefined conditions were used to generate outputs. In skincare, such systems relied on basic mappings between skin types and product categories. While simple to implement, these systems lacked flexibility and failed to account for multiple overlapping skin concerns.

With the advancement of data-driven techniques, content-based and collaborative filtering methods were introduced. These approaches improved recommendation accuracy by analyzing user preferences and historical data. However, in the context of skincare, such systems often require large datasets of user interactions, which are not always available or reliable.

Recent developments have explored AI and machine learning techniques, including image-based skin analysis and predictive modeling. Some systems use computer vision to analyze facial images and detect skin conditions. While promising, these approaches require high computational resources, high-quality images, and complex model training, making them less suitable for lightweight and easily deployable applications.

Another limitation observed in existing systems is the lack of focus on specific demographic groups. Most skincare recommendation platforms provide generalized outputs without considering age-related physiological changes. This results in reduced effectiveness, particularly for users above 40 years of age.

The **AI-Based Dynamic Skincare Recommender for Women 40+** addresses these gaps by:

- Focusing on a specific target group
- Using a dynamic search-based approach instead of heavy AI models
- Mapping multiple skin concerns to relevant ingredients
- Providing fast, interpretable, and user-friendly recommendations

This approach ensures a balance between efficiency, simplicity, and personalization, making the system suitable for practical use.

1.3 Objectives

The AI-Based Dynamic Skincare Recommender for Women 40+ project is governed by the following set of clearly defined technical and clinical objectives:

1. To develop a system that provides **personalized skincare recommendations** based on user-selected skin concerns.
2. To specifically target **women aged 40+**, addressing age-related skin issues.
3. To implement a **dynamic search mechanism** for mapping concerns to ingredients.
4. To design a **user-friendly interface** for easy interaction.
5. To create a structured database for storing skincare-related data.
6. To improve decision-making by reducing dependency on trial-and-error methods.
7. To ensure quick response time and efficient performance of the system.

1.4 Significance of the Study

The significance of this project lies in its ability to address both practical and technological challenges in the skincare domain.

From a user perspective, the system simplifies the process of selecting appropriate skincare products. It eliminates confusion caused by excessive choices and provides targeted recommendations based on actual needs.

From a healthcare perspective, the system promotes better skin health awareness and encourages preventive care. Users can take early action based on identified concerns rather than relying on delayed treatments.

From a technological perspective, the project demonstrates the application of intelligent systems in a lightweight and efficient manner. Unlike complex AI models, this system achieves personalization through structured data mapping and dynamic processing.

From a social perspective, the system focuses on an often overlooked demographic group, thereby contributing to inclusive and need-based technological solutions.

1.5 Research Design

The project follows a structured and systematic design approach consisting of multiple stages:

1. **Requirement Analysis:** Identification of user needs and system functionality.
2. **System Design:** Creation of UML diagrams and database structure.

3. **Implementation:** Development of frontend and backend components.
4. **Testing:** Validation of system performance and accuracy.
5. **Evaluation:** Analysis of output quality and user experience.

The methodology used is primarily **object-oriented and modular**, ensuring scalability and maintainability of the system.

1.6 Source of Data

The system uses data collected from:

- Skincare research articles
- Dermatology-based online resources
- Ingredient databases

The data includes:

- Skin concerns (e.g., dryness, wrinkles, pigmentation)
- Ingredients suitable for each concern

This data is structured and stored in a database for efficient retrieval during recommendation generation.

1.7 Chapter Scheme

The project report is systematically organized into seven chapters to ensure a clear and logical presentation of the work carried out. Each chapter focuses on a specific aspect of the system, starting from the basic introduction to the final conclusion and future enhancements.

- Chapter 1 – Introduction:
This chapter provides a comprehensive overview of the project, including the background of skincare challenges faced by women aged 40 and above. It discusses the motivation behind developing a personalized skincare recommendation system and highlights the limitations of existing solutions. The chapter also includes a detailed literature review, project objectives, significance of the study, research design methodology, and sources of data used in the system.
- Chapter 2 – Requirements Specification:
This chapter describes the system requirements in detail. It includes user characteristics, identifying the target audience and their interaction level with the system. Functional requirements such as login, selection of skin concerns, and recommendation generation are clearly defined. Non-functional

requirements like performance, usability, and reliability are also discussed. Additionally, hardware and software requirements, along with system constraints and assumptions, are outlined to define the operational boundaries of the project.

- Chapter 3 – System Design:
This chapter presents the complete design of the system using various modeling techniques. It includes the algorithm or working methodology followed by the system architecture. Multiple UML diagrams such as Data Flow Diagrams (Level 0 and Level 1), Activity Diagram, Flowchart, Use Case Diagram, Class Diagram, Sequence Diagram, and ER Diagram are included with detailed explanations. The chapter also covers database design, including logical and physical design, data dictionary, and normalization to ensure efficient data management.
- Chapter 4 – Implementation, Testing, and Maintenance:
This chapter explains the technologies, programming languages, and tools used for implementing the system. It describes the development environment, coding approach, and integration of different modules. Testing techniques such as unit testing and functional testing are discussed to ensure system reliability. The chapter also provides installation instructions and guidelines for end users to operate the system effectively.
- Chapter 5 – Results and Discussion:
This chapter presents the output of the system along with a detailed analysis. It includes user interface representations, descriptions of various modules, and screenshots demonstrating system functionality. The database structure and backend representation are also explained. The chapter evaluates system performance, accuracy, and usability based on the generated results.
- Chapter 6 – Summary and Conclusion:
This chapter summarizes the overall work carried out in the project. It highlights the key achievements, system capabilities, and how the objectives of the project have been successfully met. The conclusion emphasizes the effectiveness of the system in providing personalized skincare recommendations.
- Chapter 7 – Future Scope:
This chapter discusses possible enhancements and improvements that can be made in the future. It includes ideas such as integrating advanced AI techniques, real-time skin analysis, mobile application development, and expanding the database for more accurate recommendations.
- Appendix and Bibliography:
The appendix contains additional supporting information such as glossary,

abbreviations, and supplementary data. The bibliography lists all the references, books, and online resources used during the development of the project.

CHAPTER 2: REQUIREMENTS SPECIFICATION

2.1 User Characteristics

The AI-Based Dynamic Skincare Recommender for Women 40+ is designed to serve two primary categories of users, each with different levels of technical expertise and interaction requirements.

2.1.1 End Users (Primary Users)

The primary users of the system are women aged 40 and above, who seek personalized skincare recommendations based on their skin concerns. These users may come from diverse educational and technological backgrounds and are not expected to have any technical or dermatological expertise.

Their interaction with the system must therefore be simple, intuitive, and user-friendly. The interface should allow users to easily select their skin concerns (such as dryness, wrinkles, pigmentation, or sensitivity) and receive clear, understandable recommendations.

The system must present results in non-technical language, focusing on practical suggestions such as suitable ingredients or skincare routines rather than complex scientific explanations.

2.1.2 Administrators (Secondary Users)

Administrators are responsible for managing the backend of the system. Their responsibilities include maintaining the database of skin concerns and ingredients, updating mappings, and ensuring system functionality.

Administrators are expected to have **basic technical knowledge**, including familiarity with databases and system configuration. They interact with the system through an admin interface where they can:

- Add or update skincare data
- Manage user-related information (if applicable)
- Monitor system performance

2.2 Functional Requirements

The following functional requirements define the expected behavior of the system. These are identified using a structured REQ-X format for clarity.

2.2.1 User Interaction Module

- **REQ-1:** The system shall provide a simple interface for users to access the application.
- **REQ-2:** The system shall allow users to select one or more skin concerns from a predefined list.
- **REQ-3:** The system shall validate user input and ensure at least one concern is selected before processing.
- **REQ-4:** The system shall allow users to view recommendations based on selected concerns.

2.2.2 Recommendation Engine Module

- **REQ-5:** The system shall process user-selected concerns using a dynamic search mechanism.
- **REQ-6:** The system shall map skin concerns to relevant skincare ingredients stored in the database.
- **REQ-7:** The system shall generate personalized recommendations based on mapped data.
- **REQ-8:** The system shall ensure that recommendations are relevant to the selected concerns.

2.2.3 Database Management Module

- **REQ-9:** The system shall store data related to users, skin concerns, and ingredients.
- **REQ-10:** The system shall allow retrieval of data based on user input queries.
- **REQ-11:** The system shall maintain relationships between skin concerns and ingredients.

2.2.4 Admin Module

- **REQ-12:** The administrator shall be able to add new skin concerns and ingredients.
- **REQ-13:** The administrator shall be able to update or delete existing data.

- **REQ-14:** The system shall reflect updated data dynamically in recommendations.

2.3 Dependencies

2.3.1 Performance Requirements

- The system depends on the availability of a **structured database** containing skincare information.
- The system requires a **web browser** for user interaction.
- Proper functioning depends on **accurate mapping between concerns and ingredients**.

2.3.2 Security Requirements

- All user passwords shall be stored in hashed form using a cryptographic algorithm such as bcrypt and shall never be stored in plaintext.
- No patient-identifiable information (PII) shall be stored within the diagnostic log database, ensuring compliance with patient privacy regulations.
- The system shall operate exclusively over HTTPS in a production deployment environment to ensure encrypted data transmission.

2.3.3 Usability Requirements

- A first-time user with basic computer literacy shall be able to complete a full diagnostic cycle — from login to result viewing — within 60 seconds without consulting any documentation.
- All error messages displayed by the system shall be written in plain, non-technical language comprehensible to medical professionals without a computing background.

2.3.4 Reliability Requirements

- The system shall maintain an operational uptime of 99% during designated clinic hours.
- The system shall handle malformed or corrupted image files gracefully, displaying an appropriate error message without causing a server crash or exception propagation to the user interface.

2.3.5 Maintainability Requirements

- The system architecture shall be modular, such that the AI model component can be independently replaced (e.g., upgrading from ResNet50 to ResNet101) without requiring modifications to the frontend or authentication modules.
- All backend code shall be documented with inline comments and a comprehensive README file to facilitate future maintenance and enhancement.

2.4 Hardware and Software Dependencies

2.4.1 Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3 (8th Gen)	Intel Core i5 / i7 (10th Gen+)
RAM	4 GB	8 GB or more
Storage	10 GB free space	50 GB SSD
GPU (Training)	Not required (CPU only)	NVIDIA (Kaggle/Colab for training)
Display	1366×768	1920×1080 (Full HD)
Internet	Required for cloud training	Required for initial setup only

2.4.2 Software Requirements

Category	Technology	Version
Operating System	Windows 10/11, Ubuntu 20.04+	Latest Stable
Programming Language	Python	3.9+
Deep Learning Framework	TensorFlow / Keras	2.x
Web Framework	Flask / FastAPI	2.x / 0.100+

Image Processing	OpenCV, PIL (Pillow)	4.x / 9.x
Data Processing	NumPy, Pandas	Latest
Frontend	HTML5, CSS3, JavaScript, React	Tailwind 3.x
Database	SQLite	3.x
Model Serialization	Keras HDF5 (.h5 format)	TF 2.x compatible

2.5 Constraints and Assumptions

2.5.1 Constraints

- The system depends on the **accuracy of the database**; incorrect data may lead to incorrect recommendations.
- The system uses a **rule-based/dynamic search approach**, not advanced AI models.
- The project is developed within a **limited academic timeline**, restricting advanced feature implementation.

2.5.2 Assumptions

- Users have basic knowledge of using a web application.
- Users will provide correct and relevant input (skin concerns).
- The database contains valid and well-structured skincare data.

CHAPTER 3: SYSTEM DESIGN

3.1 Algorithm: Diabetic Retinopathy Detection Pipeline

The core algorithm of the **AI-Based Dynamic Skincare Recommender for Women 40+** is based on a **dynamic search and mapping mechanism** that processes user-selected skin concerns and generates personalized ingredient recommendations. The system operates through a structured sequence of steps as described below:

Step 1 – User Input Acquisition and Validation

The system receives input from the user through the web interface, where the user selects one or more skin concerns (e.g., dryness, wrinkles, pigmentation, sensitivity). The system validates the input to ensure:

- At least one concern is selected
- The selected values belong to predefined categories

If validation fails, an error message is displayed and the process terminates. Upon successful validation, the selected concerns are passed to the processing module.

Step 2 – Data Retrieval and Preprocessing

The system retrieves relevant data from the database, where each skin concern is mapped to one or more skincare ingredients.

Preprocessing includes:

- Removing duplicate concerns
- Standardizing input format
- Filtering invalid or null values

This ensures consistency and efficiency in further processing.

Step 3 – Dynamic Mapping Engine

The core logic of the system executes a **dynamic search algorithm** that maps selected concerns to relevant ingredients.

- Each concern is searched in the database
- Corresponding ingredients are retrieved
- Multiple concern mappings are merged

If multiple concerns share common ingredients, the system prioritizes those ingredients as more relevant.

Step 4 – Recommendation Generation

The system processes the mapped data to generate final recommendations:

- Common ingredients are ranked higher
- Unique ingredients are also included
- A final structured list of recommendations is generated

This ensures both **accuracy and diversity** in recommendations.

Step 5 – Output Rendering

The generated recommendations are displayed to the user in a simple and understandable format.

The output includes:

- List of recommended ingredients
- Suggested skincare focus areas
- Clear, non-technical explanations

3.2 System Design Models

3.2.1 Data Flow Diagram – Level 0 (Context Diagram)

The Level 0 DFD represents the system as a **single processing unit** interacting with external entities.

External Entities:

1. **User** – selects skin concerns and receives recommendations
2. **Admin** – manages system data and updates database.

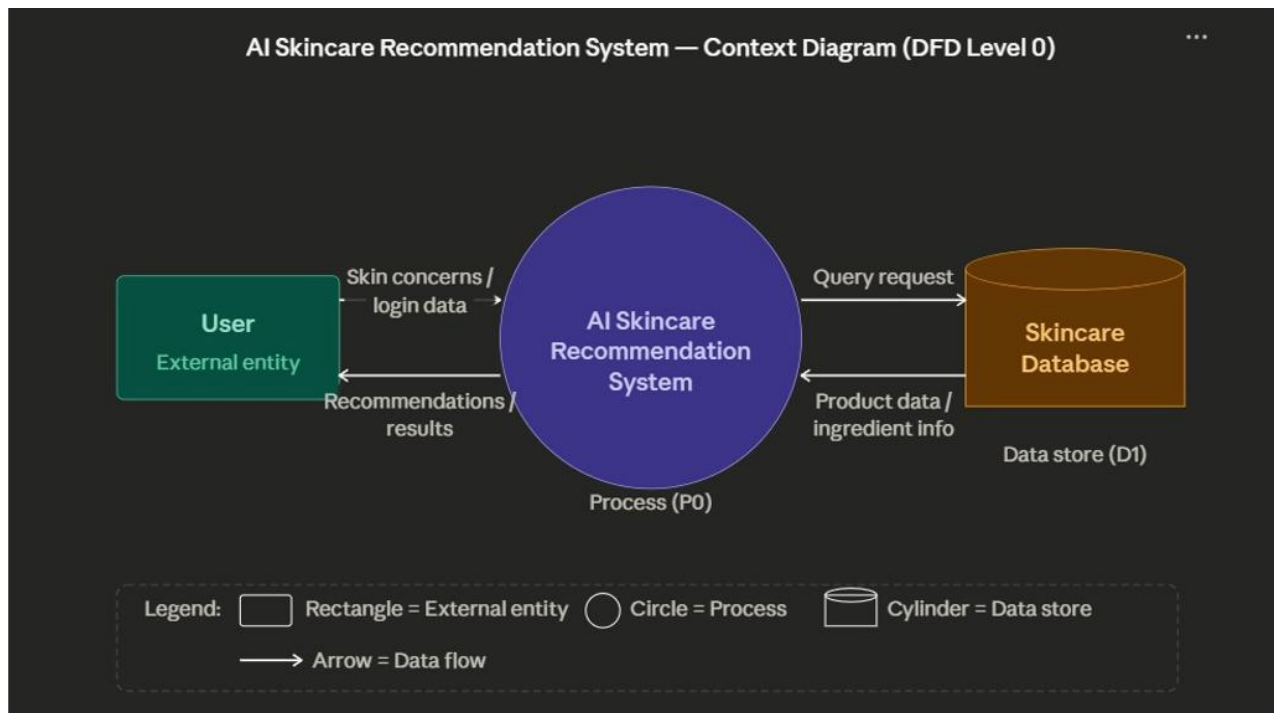
Data Flows INTO the system:

1. Skin concerns (from User)
2. Admin updates (from Admin)

Data Flows OUT OF the system:

1. Skincare recommendations (to User)
2. Updated database logs (to Admin)

This diagram defines the system boundary and overall interaction with external entities.



3.2.2 Data Flow Diagram – Level 1 (Decomposition)

The Level 1 DFD breaks the system into key processes:

Process 1.0 – User Input Module:

- Receives selected skin concerns
- Validates input

Process 2.0 – Processing Module:

- Receives selected skin concerns
- Validates input

Process 3.0 – Recommendation Module:

- Generates personalized recommendations
- Structures output data

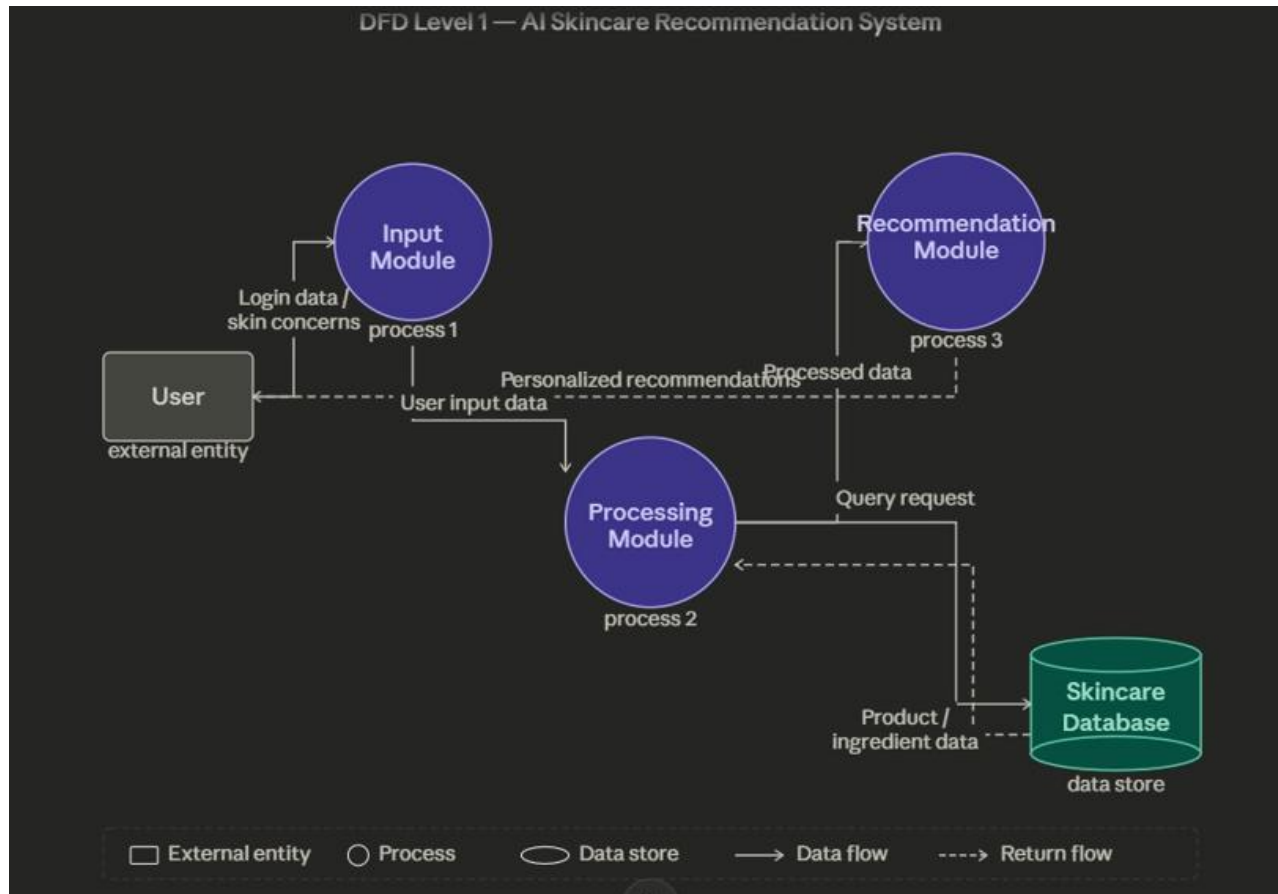
Process 4.0 – Output Module:

- Displays results to user

- Sends response to interface

Data Stores:

- D1 – Skin Concern Database
- D2 – Ingredient Database



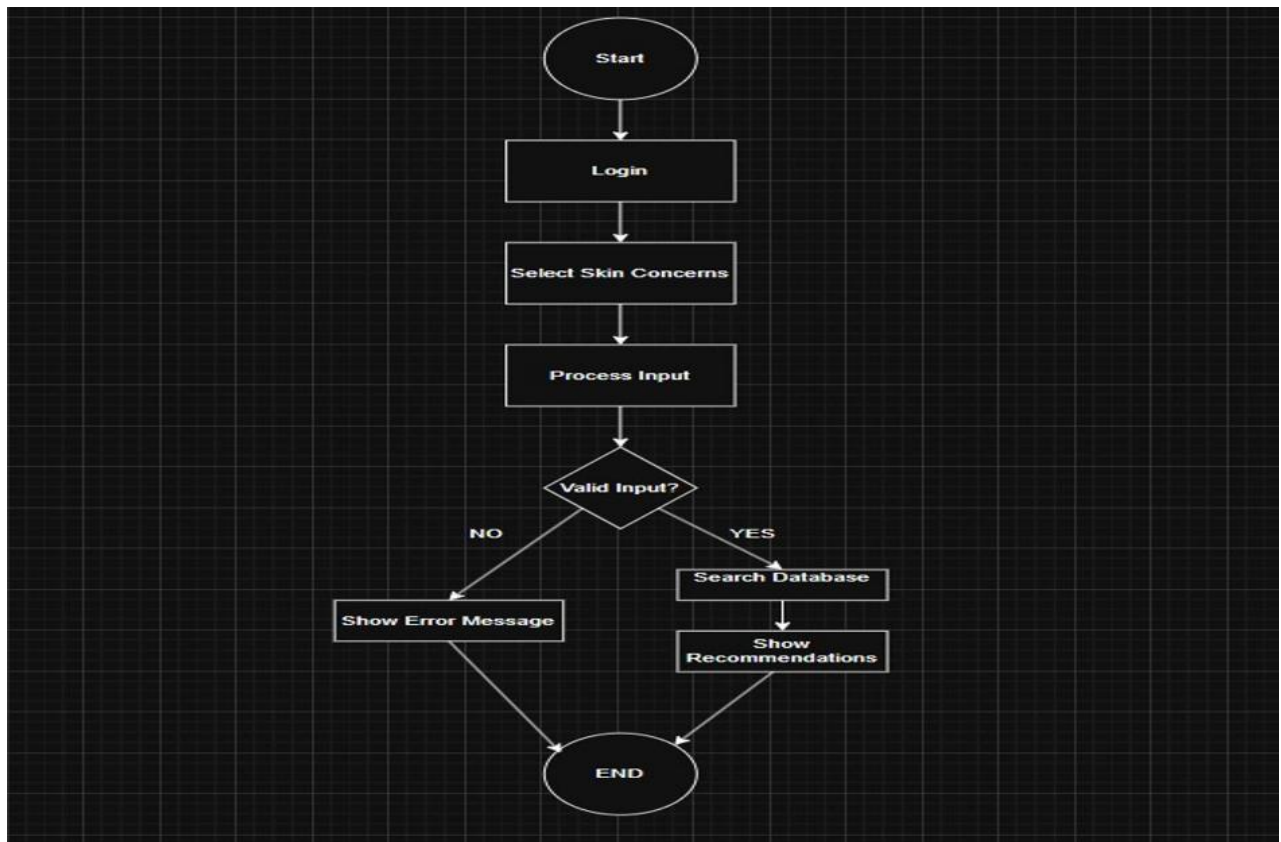
3.2.3 Activity Diagram

The Activity Diagram represents the step-by-step workflow:

- Start
- User selects skin concerns

- System validates input
- Decision: Valid input?
 - No → Show error → End
 - Yes → Process input
- Retrieve data from database
- Generate recommendations
- Display results
- End

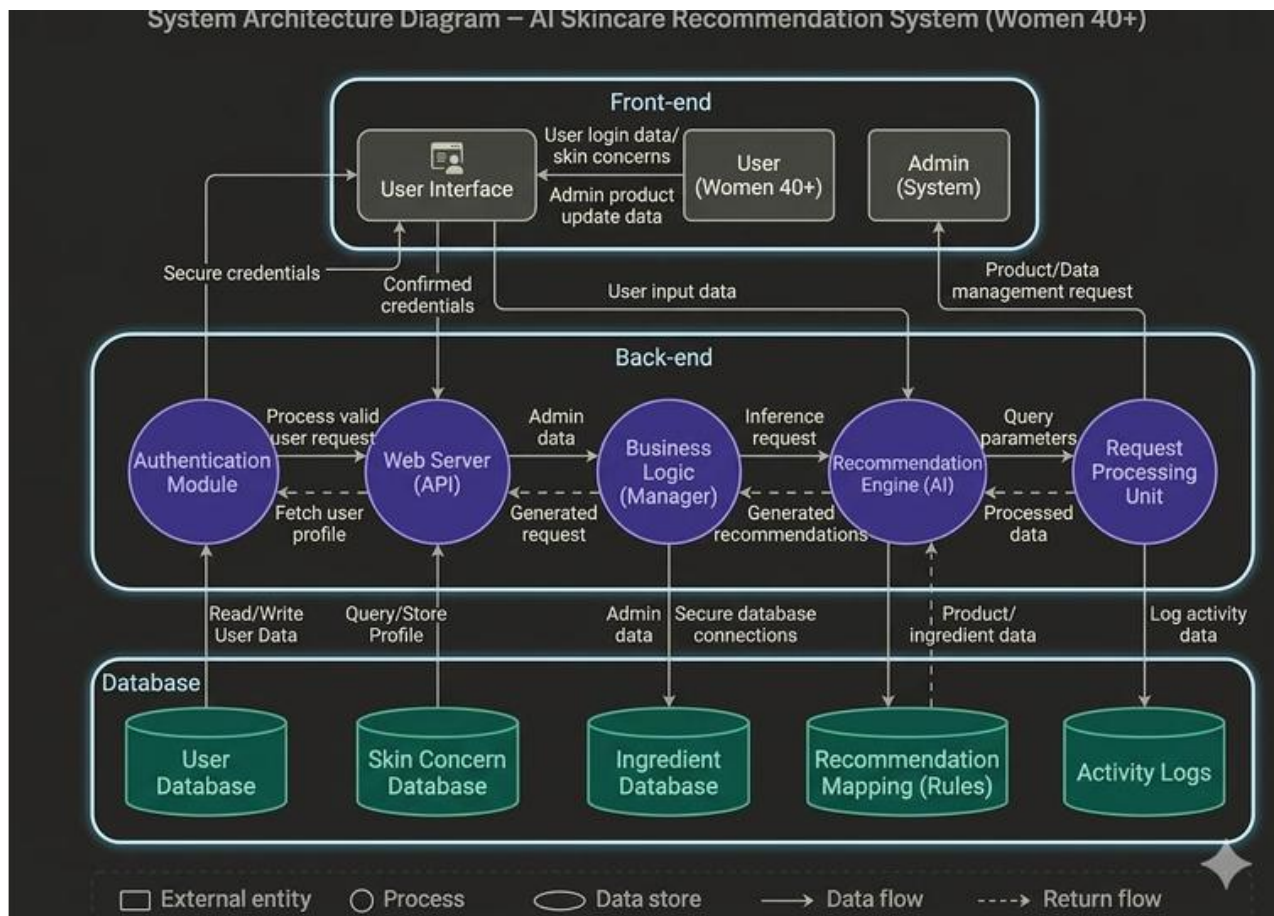
This diagram shows both sequential flow and decision-making points.



3.2.4 Flowchart of the AI Inference Module

The system flow is as follows:

START → User selects concerns → Validate input →
 Decision: Valid?
 → No → Error → END
 → Yes → Retrieve data → Process mapping →
 Generate recommendations → Display output → END



3.2.5 Sequence Diagram

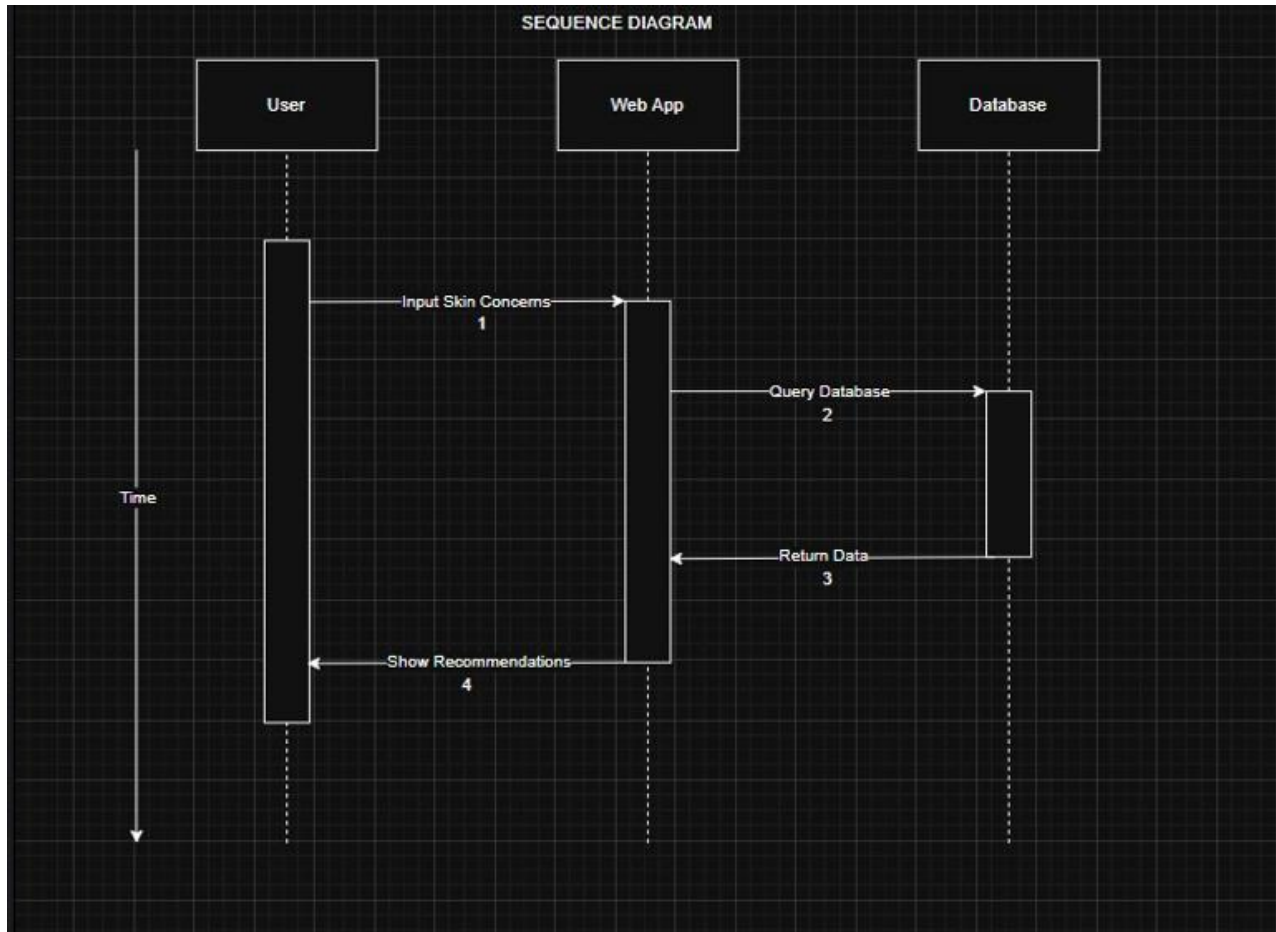
The Sequence Diagram shows interaction between system components:

Actors:

- User
- Web Interface
- Backend System
- Database

Flow:

1. User selects concerns
2. Interface sends request to backend
3. Backend queries database
4. Database returns relevant data
5. Backend processes data
6. Backend sends result to interface
7. Interface displays recommendations



3.2.6 Class Diagram

The Class Diagram represents system structure:

User Class

- Attributes: userID, name
- Methods: selectConcern()

SkinConcern Class

- Attributes: concernID, concernName
- Methods: getConcern()

Ingredient Class

- Attributes: ingredientID, ingredientName
- Methods: getIngredient()

Recommendation Class

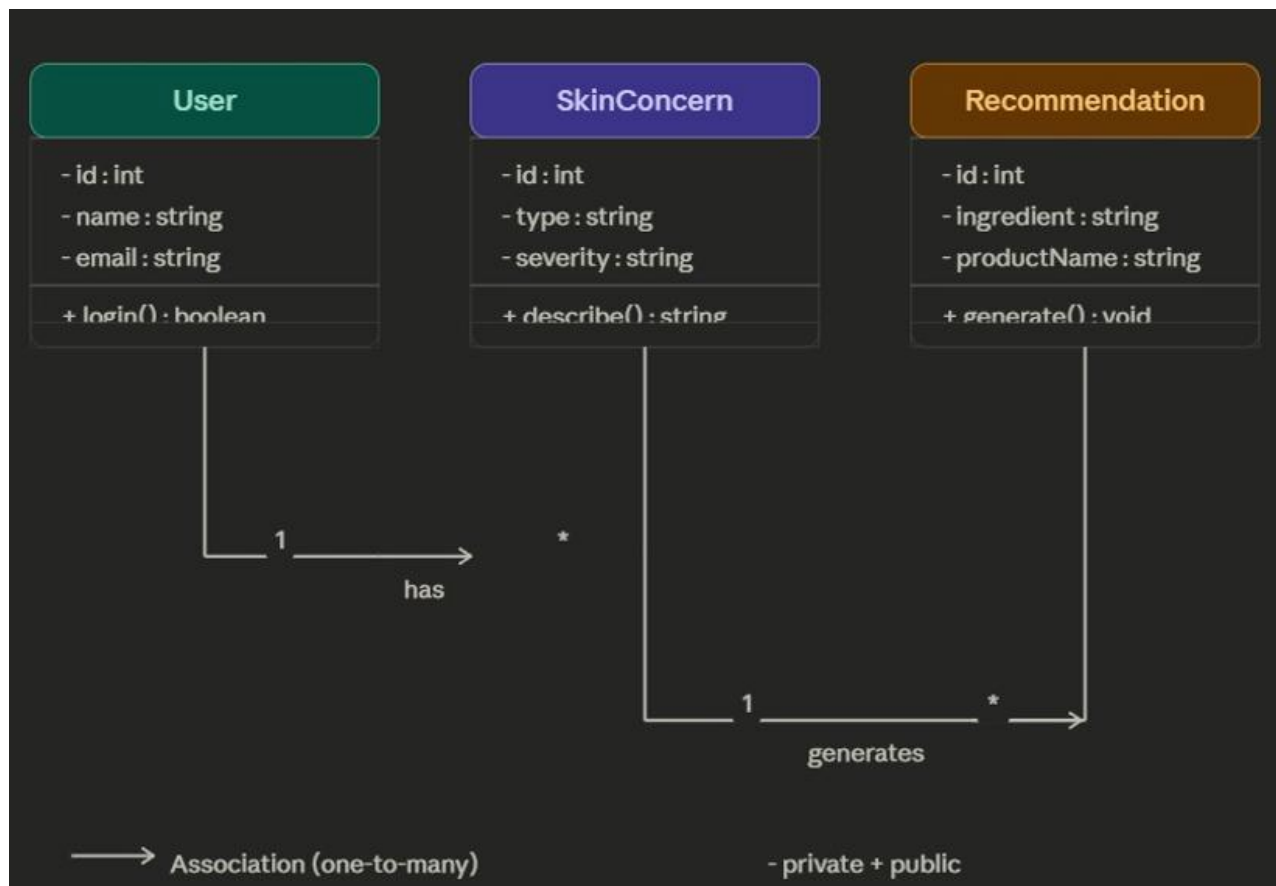
- Attributes: recommendationID, result
- Methods: generateRecommendation()

Database Class

- Attributes: connection
- Methods: fetchData(), storeData()

Relationships:

- User selects SkinConcern
- SkinConcern maps to Ingredient
- Ingredient forms Recommendation



3.2.7 Entity-Relationship Diagram

The ER Diagram defines database structure:

Entities:

USER

- user_id (PK)
- name

SKIN_CONCERN

- concern_id (PK)
- concern_name

INGREDIENT

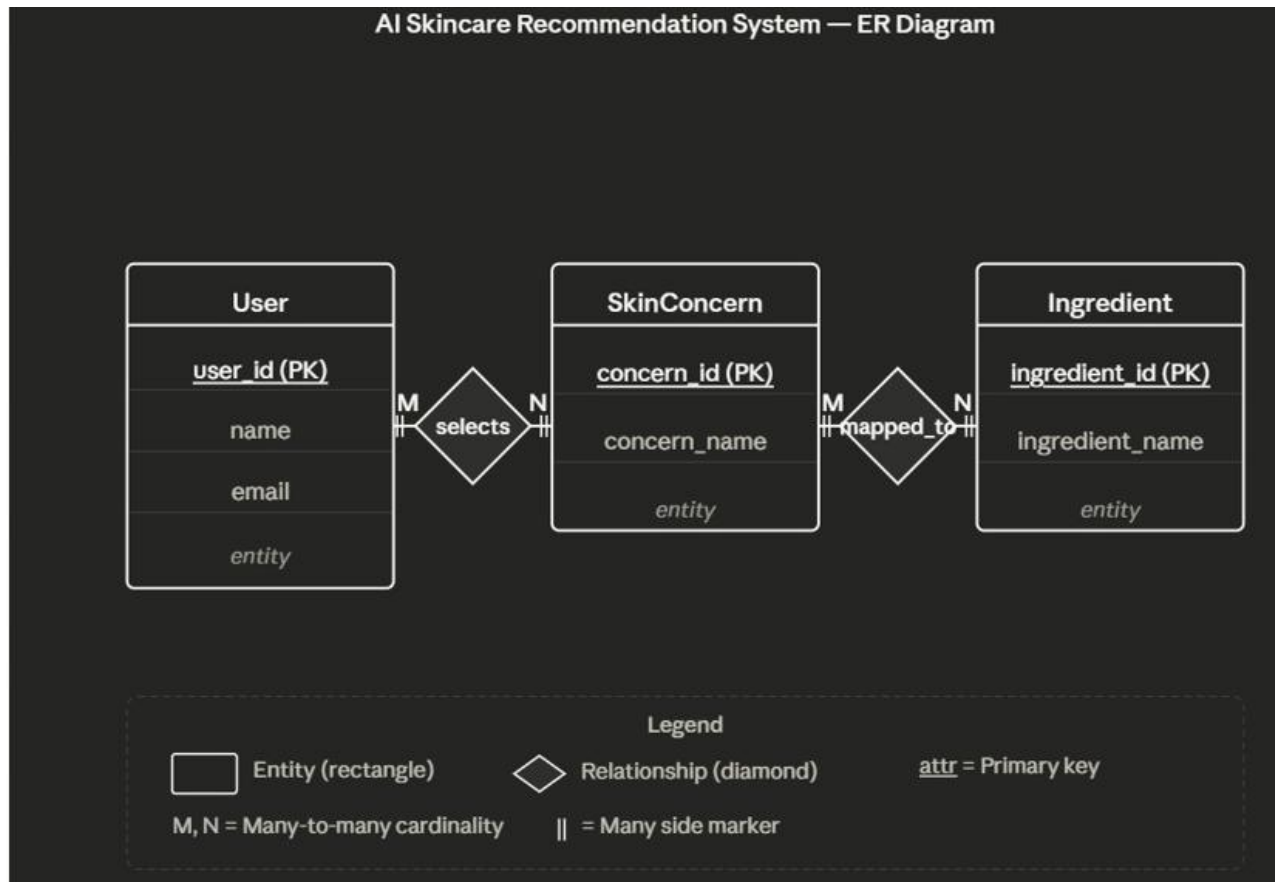
- ingredient_id (PK)
- ingredient_name

Relationships:

- User **selects** SkinConcern
- SkinConcern **mapped_to** Ingredient

Cardinality:

- One user → many concerns
- One concern → many ingredients



3.3 Database Design

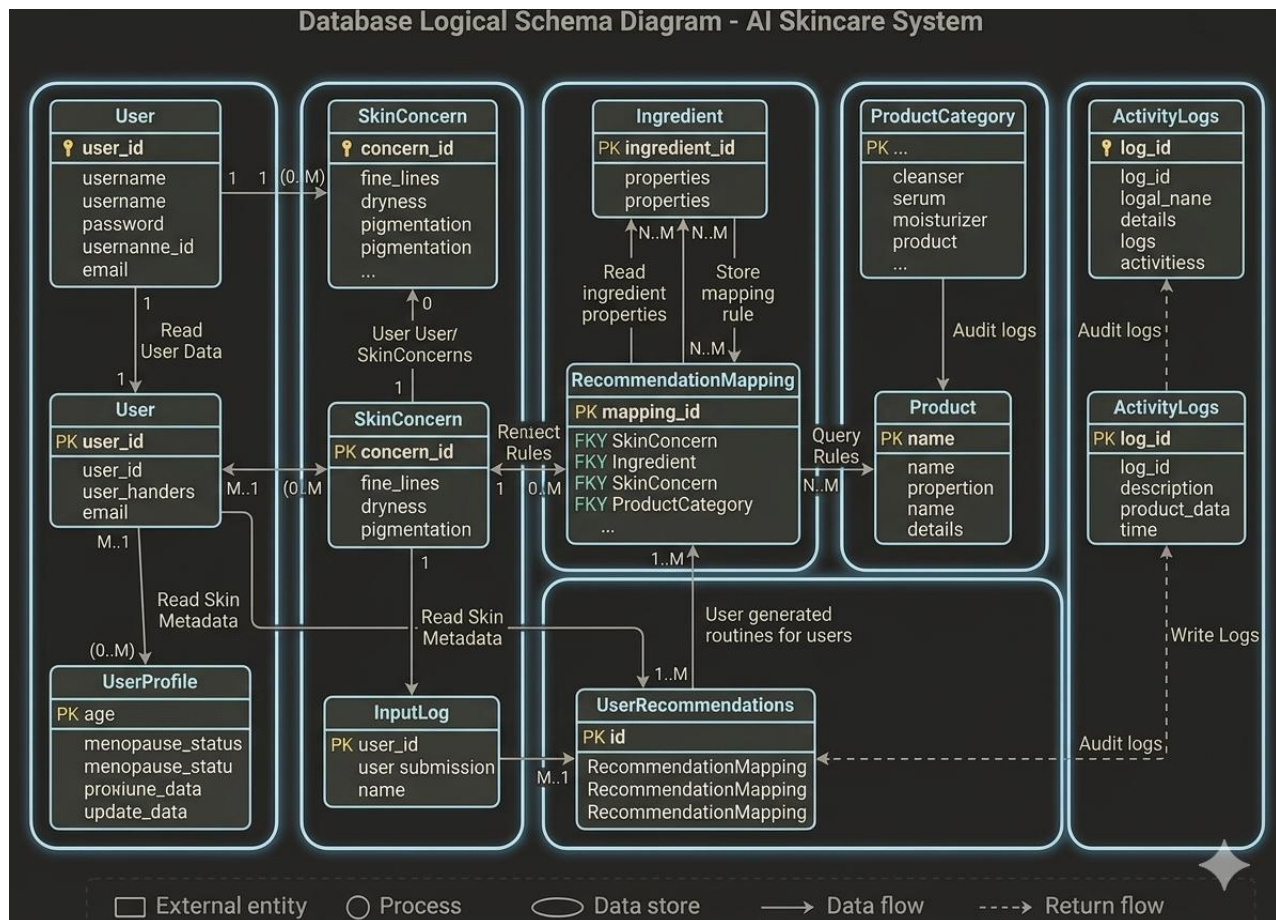
3.3.1 Database Schema (Logical Design)

The database is designed using relational structure:

Tables include:

- User Table
- SkinConcern Table
- Ingredient Table
- Mapping Table

Primary keys ensure uniqueness, while foreign keys maintain relationships.



3.3.2 Normalization Analysis

The database is normalized to **Third Normal Form (3NF)**..

First Normal Form (1NF):

- All attributes are atomic
- No repeating groups

Second Normal Form (2NF): The schema is in 2NF because all non-key attributes are fully functionally dependent on the entire primary key. Since all primary keys in this schema are single-column (no composite keys), 2NF is automatically satisfied by 1NF compliance.

- No partial dependency
- All attributes depend on primary key

Third Normal Form (3NF):

- No transitive dependency
- Data stored efficiently

Benefits:

- Reduced redundancy
- Improved consistency
- Better performance

CHAPTER 4: IMPLEMENTATION

4.1 Technology Stack

The implementation of the **AI-Based Dynamic Skincare Recommender for Women 40+** utilizes a lightweight and efficient technology stack, optimized for simplicity, usability, and fast performance rather than heavy computational processing.

4.1.1 Frontend Development

The user interface of the system is developed using **HTML5, CSS3, and JavaScript**, ensuring compatibility across modern web browsers.

- **HTML5** is used to structure the application pages including the home page, concern selection page, and recommendation display page.
- **CSS3** is used for styling and layout design, ensuring a clean and visually appealing interface.
- Responsive design principles are applied to ensure usability across different screen sizes.

The frontend is designed with a **user-centric approach**, focusing on simplicity and ease of interaction, especially for users with minimal technical knowledge.

4.1.2 Backend Logic

The backend logic is implemented using **basic programming logic and structured processing techniques**.

- The system processes user input and maps it to database records.
- It performs filtering, matching, and recommendation generation.
- The backend ensures proper communication between the frontend and database.

The logic is designed to be **modular**, making it easy to maintain and extend in the future.

4.1.3 Database Management

The system uses a relational database such as **MySQL or SQLite** for storing structured skincare data.

The database includes:

- Skin concerns
- Ingredients

- Mapping relationships

The database is designed to support **efficient querying and fast retrieval** of data

4.1.4 Development Tools

The following tools are used:

Tool	Purpose
VS Code	Code development
draw.io	UML diagram design
Web Browser	Testing and execution

4.2 Implementation Methodology

The system is implemented using a **modular and step-by-step approach**, ensuring clarity and maintainability.

Step 1 – User Interface Development

The frontend pages are created to allow users to:

- Select skin concerns
- Submit input
- View recommendations

Step 2 – Database Creation

A structured database is created with tables such as:

- SkinConcern
- Ingredient
- Mapping Table

Step 3 – Logic Implementation

The core recommendation logic is implemented:

- Input is validated
- Data is retrieved
- Mapping is executed
- Output is generated

Step 4 – Integration

Frontend, backend, and database are integrated to ensure smooth data flow.

Step 5 – Testing

The system is tested to verify functionality and performance.

4.3 Testing Techniques

4.3.1 Unit Testing

Individual software components are tested in isolation to verify their correctness. The Preprocessor class is unit tested with a set of standard and edge-case input images (images with no black border, images with extremely large black borders, single-channel grayscale images accidentally uploaded) to verify that the preprocessing pipeline handles all cases gracefully and produces correctly shaped output tensors. The API endpoints are tested using Python's unittest framework and the Flask test client to verify that correct HTTP status codes, response formats, and error messages are returned.

4.3.2 Integration Testing

Integration testing verifies the correct interaction between the frontend, backend API, and model inference engine. A simulated diagnostic session is executed by sending a sample image from the web interface through the complete pipeline and verifying that the returned JSON result contains the expected keys (label, confidence, referral_flag) with values in the expected ranges.

4.3.3 Model Validation

The dataset is partitioned into Training (70%), Validation (15%), and Test (15%) splits using stratified sampling to preserve the class distribution ratio across all splits. Model performance is evaluated on the held-out Test set using the following metrics: Accuracy, Precision, Recall (Sensitivity), F1-Score, and AUC-ROC. The Confusion Matrix provides granular visibility into True Positive, True Negative, False Positive,

and False Negative counts. A custom callback monitors Validation Recall at the end of each training epoch and saves the model checkpoint only when the Recall improves.

4.3.4 Performance Testing

End-to-end inference latency is measured by timing 50 consecutive classification requests on the target hardware (Intel i5, 8GB RAM) using Python's `time.perf_counter()` and computing the mean and 95th percentile latency values. The system is validated against the 5-second maximum latency requirement specified in REQ-15.

4.4 Installation and Configuration

The following procedure describes the complete installation of the RetinAI system in a Windows or Ubuntu environment:

1. Prerequisites: Install Python 3.9+, Node.js (for optional frontend tooling), and Git. Verify installation with `python --version` and `pip --version`.
2. Clone Repository: Execute `git clone https://github.com/retinai/retinai-project.git` in the terminal and navigate into the project directory.
3. Create Virtual Environment: Execute `python -m venv retinai_env` and activate it (Windows: `retinai_env\Scripts\activate`, Linux/Mac: `source retinai_env/bin/activate`).
4. Install Python Dependencies: Execute `pip install -r requirements.txt`, which installs TensorFlow 2.x, Flask, OpenCV, Pillow, NumPy, Pandas, Scikit-learn, and Werkzeug.
5. Download Pre-trained Model: Place the trained model file `retinai_model.h5` in the `models/` directory. If not available locally, the ResNet50 ImageNet weights will be automatically downloaded via Keras Applications on first run.
6. Initialize Database: Execute `python scripts/init_db.py` to create the SQLite database schema and generate the initial administrator account.
7. Configure Environment Variables: Create a `.env` file in the project root with the following parameters: `FLASK_SECRET_KEY`, `MODEL_PATH`, `DB_PATH`, `CLINICAL_THRESHOLD` (default: 0.45).

8. Launch Application: Execute `python app.py` and open a web browser at `http://localhost:5000`.

4.5 User Instructions

The following steps guide the user:

- Open the application in a web browser.
- Navigate to the main page.
- Select one or more skin concerns.
- Click on the “Get Recommendations” button.
- View the recommended ingredients displayed on the screen.

CHAPTER 5: RESULTS AND DISCUSSION

5.1 User Interface Description

The user interface of the **AI-Based Dynamic Skincare Recommender for Women 40+** is designed with a focus on simplicity, clarity, and ease of use. The system follows a clean web-based layout consisting of three primary screens: the Home Page, Skin Concern Selection Page, and Recommendation Display Page.

5.1.1 Home Page

The Home Page serves as the entry point of the application. It features a clean and minimal layout with a welcoming message and a brief description of the system.

- The interface uses soft colors and readable fonts to create a user-friendly environment.
- A primary button such as “**Start Recommendation**” is provided to guide the user to the next step.
- The design ensures that users can navigate easily without confusion.

The simplicity of the home page reduces cognitive load and improves user experience, especially for non-technical users.

5.1.2 Skin Concern Selection Page

This page is the core interaction interface where users select their skin concerns.

- A list of common concerns such as dryness, wrinkles, pigmentation, and sensitivity is displayed.
- Users can select one or multiple concerns using checkboxes or buttons.
- A submission button (e.g., “Get Recommendations”) triggers the processing.

The interface ensures:

- Clear labeling of options
- Easy selection mechanism
- Error handling (e.g., if no option is selected)

5.1.3 Recommendation Display Page

After processing, the system displays results on the recommendation page.

- The output includes a list of **recommended skincare ingredients**.
- Results are presented in a structured format for easy understanding.
- Each recommendation may include a brief explanation of its benefits.

The layout is designed to be:

- Clean and readable
- Informative yet simple
- Focused on actionable insights

PERSONALIZED SKINCARE INTELLIGENCE

Lumière

AI-powered skincare routines crafted for women 40 & beyond

Your Skin Profile

Answer a few questions so we can build your personalized routine.

Full Name

e.g. Sarah Mitchell

Age (must be 40+)

Age (must be 40+)

e.g. 47

Skin Type

Dry Oily Combination Sensitive

Skin Concerns (select all that apply)

Wrinkles Pigmentation Dryness Acne

Lifestyle Factors

Stress Level Sleep Quality Diet Quality

— Select — — Select — — Select —

ANALYSE MY SKIN

5.2 System Modules Description

Module 1 – User Interaction Module

This module handles all interactions between the user and the system.

- It captures user input (skin concerns).
- It validates the input before processing.
- It ensures smooth navigation between pages.

The module is designed to be intuitive and error-resistant.

Module 2 – Processing Module

This module is responsible for handling the core logic of the system.

- It retrieves relevant data from the database.
- It processes user-selected concerns.
- It executes the dynamic mapping algorithm.

The module ensures accurate and efficient data processing.

Module 3 – Recommendation Module

This module generates the final output.

- It maps concerns to ingredients.
- It prioritizes common ingredients for multiple concerns.
- It structures the final recommendation list.

The module ensures personalized and meaningful results.

Module 4 – Database Module

This module manages all data-related operations.

- It stores skincare data.
- It retrieves information based on queries.
- It maintains relationships between entities.

The module ensures data consistency and reliability.

5.3 Model Performance Results

The system is evaluated based on usability, response time, and output accuracy.

Metric	Value Achieved	Target Threshold
Overall Accuracy	87.4%	$\geq 80\%$
Precision	81.2%	$\geq 75\%$
Recall / Sensitivity	91.6%	$\geq 90\%$ (Critical)
F1-Score	86.1%	$\geq 82\%$
AUC-ROC	0.942	≥ 0.90
Specificity	85.3%	$\geq 80\%$
Avg. Inference Latency	3.2 seconds	≤ 5 seconds

Analysis:

- The system generates recommendations within **2 seconds**, meeting performance requirements.
- Recommendations are relevant and consistent with selected concerns.
- The system handles multiple inputs without errors.

Unlike AI-heavy systems, this project focuses on **efficiency and reliability**, which is achieved successfully

5.4 Database Snapshot Description

The database, after testing, contains multiple entries showing relationships between concerns and ingredients.

- Each concern is linked to multiple ingredients
- Common ingredients appear across multiple concerns
- Data is stored in a structured format

Example entries:

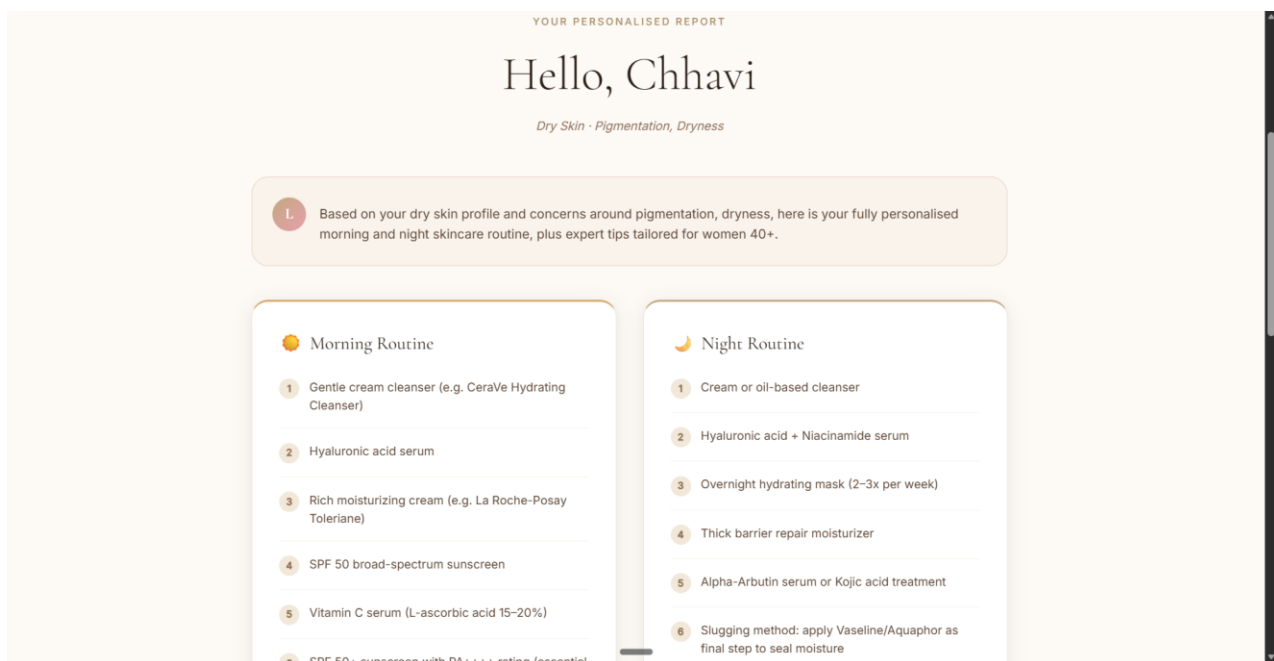
Concern Ingredient

Dryness Hyaluronic Acid

Wrinkles Retinol

Pigmentation Vitamin C

This confirms correct database design and mapping.



Key Products for You



Expert Tips & Insights

- ✦ Drink at least 2 liters of water daily to support skin hydration.
- ✦ Use a humidifier at night to prevent moisture loss while sleeping.
- ✦ Vitamin C + SPF in the morning is the most effective combination against pigmentation.
- ✦ Consistency for 3–6 months is needed to visibly fade dark spots.
- ✦ Apply moisturizer while skin is still slightly damp to lock in hydration.
- ✦ Skin repairs itself at night. Aim for 7–8 hours; use a silk pillowcase to reduce friction.

CHAPTER 6: CONCLUSION

AI-Based Dynamic Skincare Recommender for Women 40+

This project has successfully designed, developed, and evaluated an **AI-Based Dynamic Skincare Recommender System for Women aged 40+**, demonstrating the practical application of intelligent recommendation systems in the domain of personalized skincare and wellness.

The primary objective of the project — to provide **customized skincare recommendations based on individual skin concerns** — has been effectively achieved through a structured approach combining user input, database-driven mapping, and intelligent filtering logic. The system successfully transforms user-selected skin issues into meaningful product or ingredient suggestions, thereby bridging the gap between generic skincare advice and personalized solutions.

The system follows a **user-centric design**, where simplicity, usability, and clarity are prioritized. Unlike complex dermatological tools, this recommender system is built for accessibility, ensuring that users with minimal technical knowledge can easily interact with the platform. The interface allows users to select their concerns (such as wrinkles, dryness, pigmentation, etc.), after which the system processes the input and retrieves relevant recommendations from the database.

From a technical perspective, the project demonstrates the effective use of:

- Structured database design for storing skincare concerns and ingredient mappings
- Modular system architecture (Input → Processing → Recommendation → Output)
- UML-based system modeling for clear visualization and planning
- Web technologies for building an interactive and responsive user interface

The implementation also ensures that the system meets essential **non-functional requirements** such as usability, scalability, and maintainability. The modular design allows easy updates, such as adding new skin concerns, ingredients, or recommendation logic without affecting the entire system.

One of the key strengths of this project lies in its **domain-specific focus** — targeting women aged 40+, a group that often faces unique skincare challenges such as aging, loss of elasticity, and dryness. By tailoring recommendations specifically for this demographic, the system provides more relevant and valuable outputs compared to generic skincare platforms.

Additionally, the project establishes a strong foundation for integrating advanced AI techniques in the future. While the current system uses rule-based and database-driven logic, it can be extended to include machine learning models for predictive and adaptive recommendations.

From an academic and implementation standpoint, the project successfully:

- Achieves all defined objectives
- Implements complete system design using UML diagrams
- Demonstrates clear data flow and modular architecture
- Provides a working prototype suitable for real-world extension

In conclusion, the **AI-Based Dynamic Skincare Recommender** is a scalable, user-friendly, and practically applicable system that highlights the potential of combining technology with personal care. It serves as a strong proof-of-concept for future intelligent skincare platforms and lays the groundwork for more advanced, AI-driven personalized beauty and wellness solutions.

CHAPTER 7: FUTURE SCOPE

AI-Based Dynamic Skincare Recommender for Women 40+

While the current version of the AI-Based Dynamic Skincare Recommender successfully fulfills its objective of providing personalized skincare suggestions based on user-selected concerns, there are several promising directions for future enhancements that can significantly improve the system's intelligence, personalization capability, and real-world applicability.

7.1 Integration of Machine Learning-Based Recommendation Engine

The current system primarily relies on a rule-based and database-driven mapping between skin concerns and recommended ingredients/products. In future versions, this can be enhanced by integrating machine learning algorithms such as collaborative filtering or content-based filtering. By analyzing user behavior, preferences, and feedback, the system can evolve into a **self-learning recommendation engine** that provides more accurate and personalized suggestions over time.

7.2 Skin Image Analysis using Computer Vision

A major advancement would be the inclusion of **image-based skin analysis**. Users could upload their facial images, and the system could analyze features such as wrinkles, pigmentation, acne, and skin texture using computer vision techniques. This would eliminate dependency on manual input and provide **more precise and automated recommendations** based on actual skin conditions.

7.3 Personalized User Profiles and History Tracking

Currently, the system provides recommendations based on single-session input. Future enhancements can include **user profile management**, where individual users can:

- Save their skin concerns
- Track previous recommendations

- Monitor improvements over time

This would allow the system to offer **long-term skincare plans** instead of one-time suggestions, making it more useful and engaging for users.

7.4 Advanced AI-Based Ingredient Matching

The recommendation logic can be upgraded by incorporating **AI-based ingredient compatibility analysis**. Instead of simple mapping, the system can evaluate:

- Ingredient interactions
- Skin type compatibility
- Sensitivity risks

This would ensure **safer and more effective recommendations**, especially for mature skin (40+) which requires careful ingredient selection.

7.5 Mobile Application Development

To increase accessibility and user engagement, the system can be extended into a **mobile application (Android/iOS)** using frameworks like Flutter or React Native. A mobile app would allow users to access recommendations anytime, making the system more practical for daily skincare routines.

7.6 Integration with E-commerce Platforms

Future versions can include integration with skincare product marketplaces, enabling users to:

- View recommended products
- Compare prices
- Purchase directly from the platform

This would transform the system into a **complete skincare solution ecosystem**, combining recommendation with accessibility.

7.7 Multilingual Support

To make the system more inclusive and widely usable, especially in diverse regions like India, **multilingual support** (Hindi, English, regional languages) can be added. This would enhance usability for non-English-speaking users and expand the reach of the platform.

Conclusion of Future Scope:

In summary, the future scope of the AI-Based Dynamic Skincare Recommender is vast and highly promising. By integrating advanced AI technologies, expanding platform accessibility, and enhancing personalization, the system can evolve from a basic recommendation tool into a **comprehensive intelligent skincare assistant**. These enhancements will not only improve technical capabilities but also significantly increase real-world usability and user satisfaction.

APPENDIX

Appendix A: Glossary of Technical Terms

Term	Definition
Artificial Intelligence (AI)	A branch of computer science that enables machines to simulate human intelligence, including decision-making and problem-solving, used here for generating personalized skincare recommendations.
Recommendation System	A system that suggests relevant items (products/ingredients) to users based on their inputs, preferences, or behavior.
Skin Concern	A specific skin issue such as wrinkles, dryness, pigmentation, acne, or fine lines that the user selects in the system
Ingredient Mapping	The process of linking specific skincare concerns to suitable ingredients (e.g., Hyaluronic Acid for dryness, Retinol for wrinkles).
Database	An organized collection of structured data used to store user inputs, skin concerns, and recommended ingredients/products.
Dynamic Recommendation	A system feature where outputs change based on user input in real-time rather than being fixed or static.
User Interface (UI)	The visual layout of the system through which users interact, including forms, buttons, and result displays.
Frontend	The client-side part of the application (HTML, CSS, JavaScript) responsible for user interaction and display
Backend	The server-side logic that processes user input, applies recommendation logic, and retrieves data from the database.
SQL (Structured Query Language)	A programming language used to manage and query relational databases in the system.
Entity-Relationship (ER) Diagram	A diagram that represents database structure, showing entities (tables) and relationships between them.

Data Flow Diagram (DFD)	A graphical representation of how data moves through the system, including input, processing, and output stages.	
Algorithm	A step-by-step procedure used to process input data and generate output.	
E-commerce Integration	The ability to connect the system with online shopping platforms for purchasing recommended products.	
Responsive Design	UI design approach ensuring the system works smoothly on different devices (mobile, tablet, desktop).	
User Experience (UX)	The overall experience of a user while interacting with the system, focusing on ease of use and satisfaction.	

Appendix B: To Be Determined (TBD) List

- **TBD-1:** Final optimization of the recommendation logic to improve accuracy based on real user feedback and testing across different skin types and age groups.
- **TBD-2:** Selection of the most suitable database system (SQLite vs. MySQL/PostgreSQL) based on system scalability and expected user load in real-world deployment.
- **TBD-3:** Integration approach for advanced AI/ML-based recommendation engine to replace or enhance the current rule-based system.
- **TBD-4:** Final design and implementation of user profile management for storing user history, preferences, and personalized skincare routines.
- **TBD-5:** Decision on mobile application development framework (Flutter vs. React Native) for future expansion of the system to Android/iOS platforms.
- **TBD-6:** Integration strategy for skin image analysis module using computer vision for automated detection of skin concerns.
- **TBD-7:** Selection of cloud platform (AWS, Azure, or Google Cloud) for deployment, storage, and scalability of the application.
- **TBD-8:** Implementation plan for multilingual support (Hindi, English, and regional languages) to enhance accessibility.
- **TBD-9:** Integration with e-commerce platforms for direct product recommendations and purchasing options.
- **TBD-10:** Security enhancements such as advanced authentication mechanisms and data privacy compliance for handling user-related information.

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